

Question:
The feedback loop in men which regulates the production of sex hormones includes all but which of the following?

- Answer Choices:
- A. follicle-stimulating hormone
 - B. oxytocin
 - C. pituitary gland
 - D. hypothalamus
 - E. gonadotropin-releasing hormone
 - F. luteinizing hormone
 - G. testosterone

Reasoning + Answer:

<think>
Concise explanation :

- Feedback loops that regulate sex hormone secretion are classical negative feedback circuits in the hypothalamic-pituitary-gonadal axis .
- Negative feedback is required for stable , pulsatile secretion of gonadotropins and thus LH and FSH .
- The hypothalamus produces GnRH , which stimulates the anterior pituitary to secrete LH and FSH . After luteinizing hormone binds to receptors on the ovary , it stimulates luteinization and , via the negative feedback loop on the pituitary via LH and FSH , returns to low levels .
- Men’s testosterone output reflects this feedback loop : pulsatile LH surges lead to follicle growth , ovulation , and testosterone production ; declining testosterone then signals pituitary LH suppression .
- Therefore , the components involved are :
- Hypothalamus via GnRH (starting the pulsatility of LH).
- Pituitary (secreting LH and FSH).
- Ovary (responding to LH and releasing testosterone).
- Negative feedback from estrogen/LH on pituitary LH secretion (the feedback loop).
- Testosterone itself is not part of the feedback loop because it does not suppress LH secretion under physiological conditions in men .

Correct answer : G . testosterone
</think>
<answer>
G . testosterone
</answer>